

EXERCISES E3

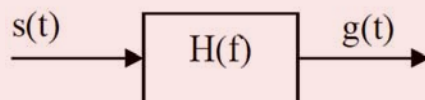
Exercise 1

A signal $s(t)$ is given whose transform is represented by the function:

$$S(f) = A \cdot \text{rect}\left(\frac{f}{B}\right) \cdot \left(2 - \frac{|f|}{B}\right)$$

where A and B are fixed constants.

The signal $s(t)$ enters a linear block with transfer function $H(f)$, and then out of the block comes the signal $g(t)$, whose Fourier transform is $G(f)$.

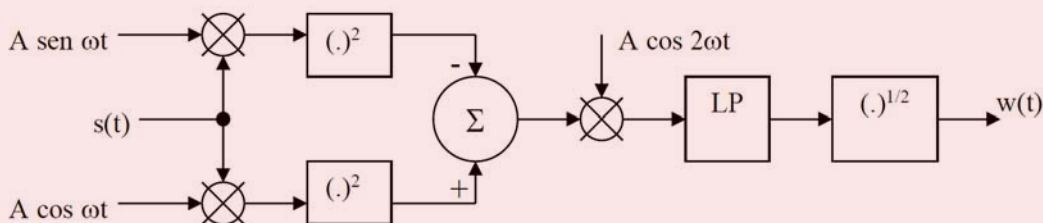


Calculate:

- The transfer function $H(f)$ so that $G(f)$ has a constant value equal to $2A$ in the frequency interval $(-B/2, +B/2)$, and zero elsewhere.
- The expression of $s(t)$ (antitransform of $S(f)$).
- The expression of $g(t)$ (antitransform of $G(f)$).

Exercise 2

The block diagram below is given:



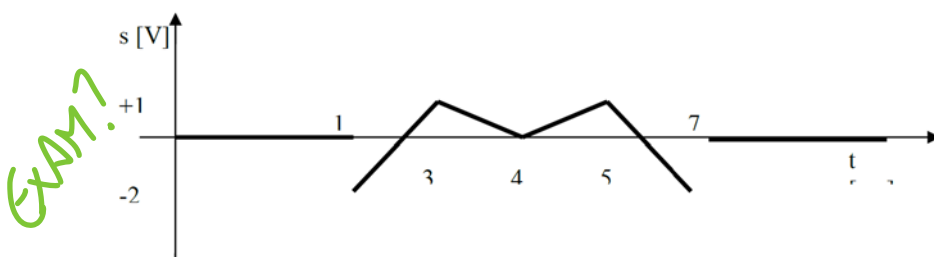
Knowing that $s(t)$ has non-zero spectrum only in the frequency interval $[0; B]$, and that $\frac{\omega}{2\pi} \gg B$, determine the expression of the signal $w(t)$.

If we can only observe the signal $w(t)$, is it possible to calculate the original $s(t)$? Why?

Given a fixed $s(t)$, what is the relationship between the value of parameter A (carrier amplitude) and the power of the signal $w(t)$?

Exercise 3

The signal $s(t)$ represented in the following graph is given:



Write the expression of $s(t)$ and its Fourier transform.

Exercise 4

Consider the two blocks in the figure.

Each block can be thought of as an approximation of a low-pass filter.

The two filters, however, have different (approximate) bandwidths.

Which of the two has the wider bandwidth?

